

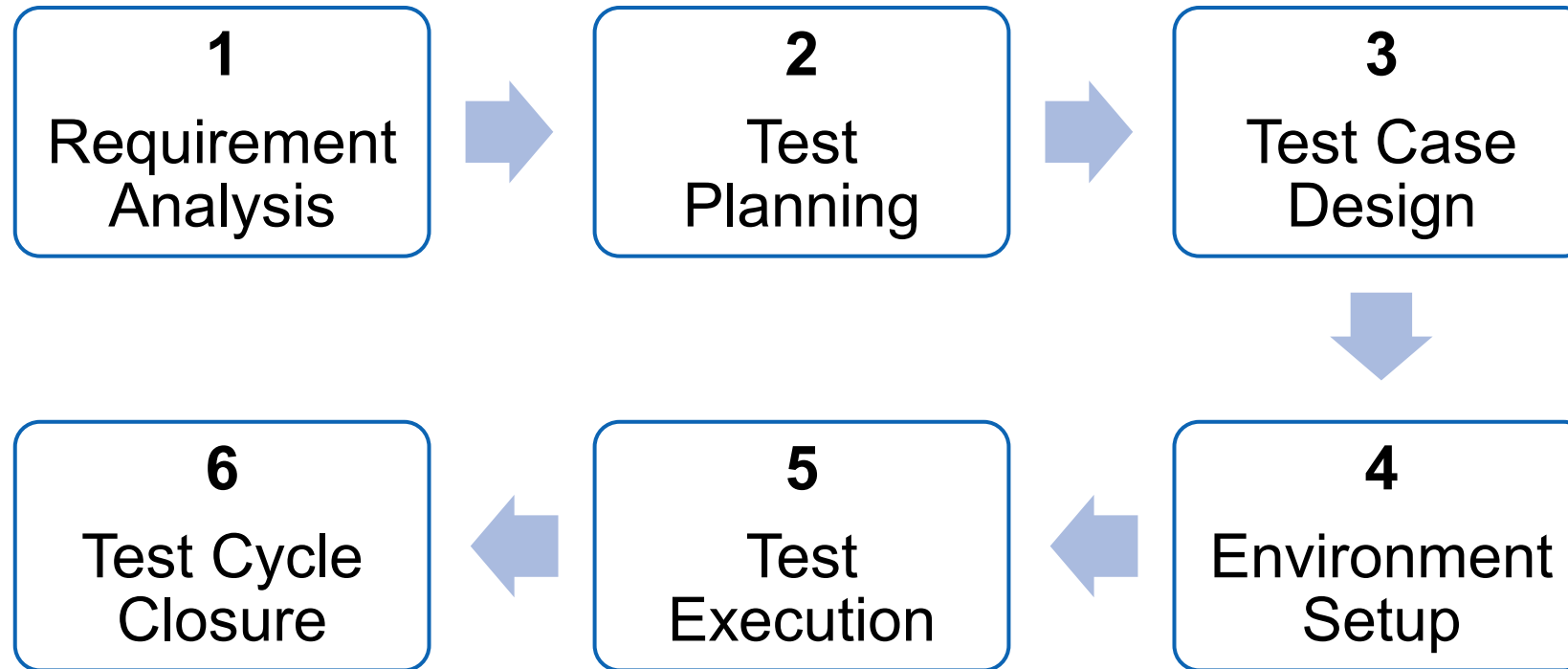
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Software Testing

CSC13003

Requirement Analysis

Software testing life cycle



Requirement analysis

- Test team
 - studies the requirements
 - from a testing point of view
 - identify testable requirements
- Interact with stakeholder
 - to better understand
- Requirements
 - functional
 - non-functional

Requirement analysis

- Activities
 - Identify types of tests to be performed.
 - Gather details about testing priorities and focus
 - Prepare Requirement Traceability Matrix
 - Identify test environment details
 - Automation feasibility analysis (if required)

Requirement analysis

- Deliverables
 - Requirement Traceability Matrix
 - Automation Feasibility Report

Requirements Traceability Matrix

- Map and trace user requirement with test cases
- Ensure that all requirements are tested (via test cases)
- Parameters
 - Requirement ID
 - Requirement description
 - Test case ID
 - Test case status

Requirements Traceability Matrix

An example of RTM

Req. ID	Req. description	Test case ID	Status
BR01	Login	TC01, TC02	TC01 – Passed TC02 – Passed
BR02	Search product	TC03, TC04, TC05	TC03 – Passed TC04 – Passed TC05 – Failed
BR03	Place an order	TC06, TC07, T08, TC09	TC06 – Passed TC07 – Passed TC08 – Failed TC09 – No run
...	...	...	...

Traceability Test Matrix

- Types of Traceability Test Matrix
 - Forward traceability
 - Map requirements to test cases
 - Make sure each requirement is tested
 - Backward or reverse traceability
 - Maps test cases to requirements
 - Make sure the scope is not expanded
 - Bi-directional traceability

Automation Feasibility Report

- Automation feasibility analysis
 - Can the application be automated or not?
 - What type of tool/ test automation framework can be used?
 - How much automation is possible?
 - In spite of high efforts, what's the value add in automating?
- Automation feasibility checklist

